4.60 A coaxial capacitor consists of two concentric, conducting, cylindrical surfaces: one of radius a and another of radius b, as shown in **Fig. P4.60**. The insulating layer separating the two conducting surfaces is divided equally into two semi-cylindrical sections: one filled with dielectric ε_1 and the other filled with dielectric ε_2.

(a) Develop an expression for C in terms of the length l and the given quantities.

*(b) Evaluate the value of C for $a = 2$ mm, $b = 6$ mm, $\varepsilon_{r_1} = 2$, $\varepsilon_{r_2} = 4$, and $l = 4$ cm.

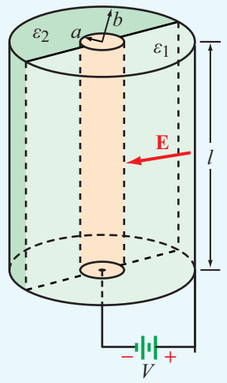


Figure P4.60 Problem 4.60.

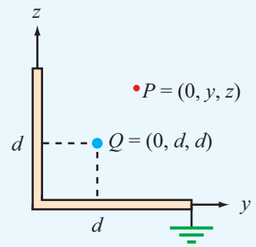


Figure P4.61 Charge Q next to two perpendicular, grounded, conducting half-planes.

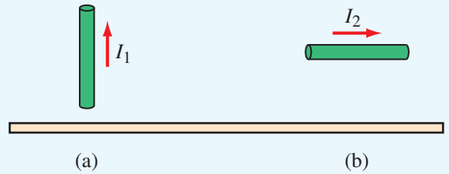


Figure P4.62 Currents above a conducting plane (Problem 4.62).

*4.63 Use the image method to find the capacitance per unit length of an infinitely long conducting cylinder of radius a situated at a distance d from a parallel conducting plane, as shown in **Fig. P4.63**.

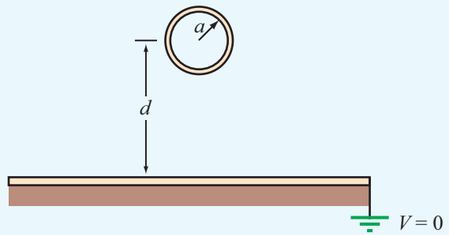


Figure P4.63 Conducting cylinder above a conducting plane (Problem 4.63).

Section 4-12: Image Method

4.61 With reference to **Fig. P4.61**, charge Q is located at a distance d above a grounded half-plane located in the x–y plane and at a distance d from another grounded half-plane in the x–z plane. Use the image method to

(a) Establish the magnitudes, polarities, and locations of the images of charge Q with respect to each of the two ground planes (as if each is infinite in extent).

(b) Find the electric potential and electric field at an arbitrary point $P = (0, y, z)$.

4.62 Conducting wires above a conducting plane carry currents I_1 and I_2 in the directions shown in **Fig. P4.62**. Keeping in mind that the direction of a current is defined in terms of the movement of positive charges, what are the directions of the image currents corresponding to I_1 and I_2?

4.60 同轴电容器由两个同心的导电圆柱形表面组成：一个半径为 a，另一个半径为 b，如图 P4.60 所示。分隔两个导电表面的绝缘层被等分为两个半圆柱形部分：一个填充有电介质ε_1，另一个填充有电介质ε_2。

(a) 根据长度 l 和给定数量建立 C 的表达式。

*(b) 计算 a = 2 mm、b = 6 mm、$\varepsilon_{r_1} = 2$、$\varepsilon_{r_2} = 4$ 和 l = 4 cm 的 C 值。

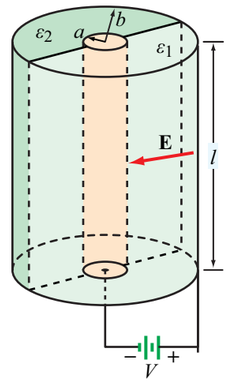


图 P4.60 问题 4.60。

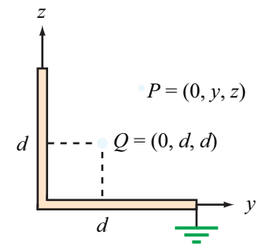


图 P4.61 在两个垂直、接地、导电的半平面旁边充电 Q。

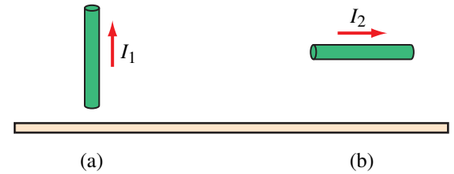


图 P4.62 导电平面上方的电流（问题 4.62）。

*4.63 用镜像法求半径为 a 的无限长导电圆柱体与平行导电面距离 d 处的单位长度电容，如图 P4.63 所示。

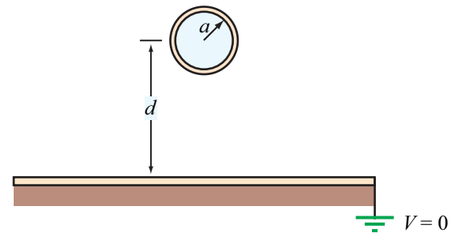


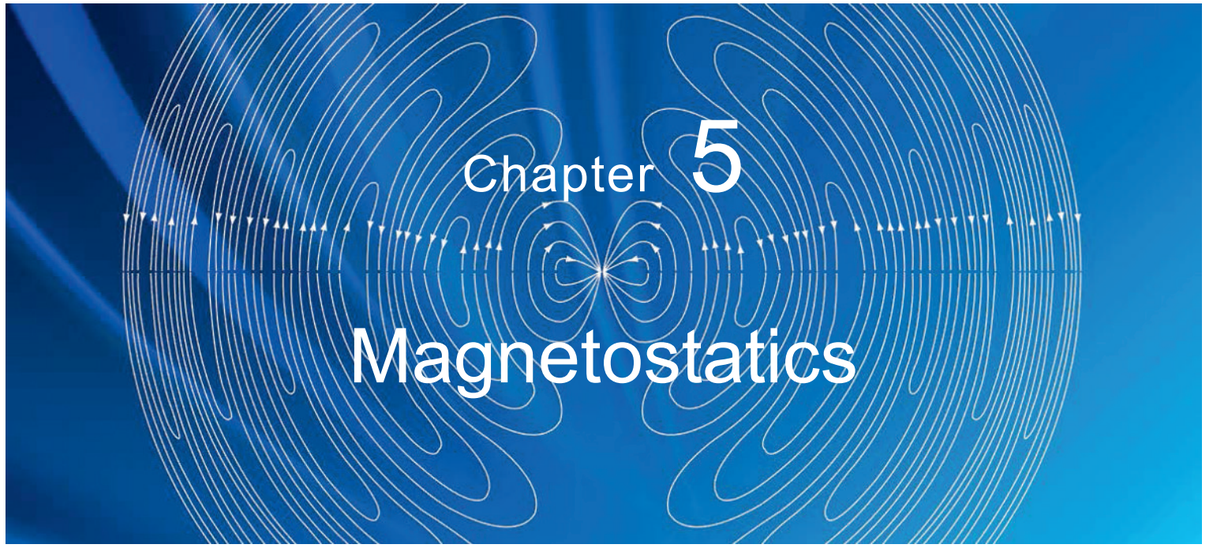
图 P4.63 导电平面上方的导电圆柱体（问题 4.63）。

第 4-12 节：镜像法

4.61 参考图 P4.61，电荷 Q 位于 x-y 平面中接地半平面上方距离 d 处，并且与 x-z 平面中另一个接地半平面距离 d 处。使用图像方法

(a) 确定电荷 Q 的图像相对于两个接地平面的大小、极性和位置（就好像每个接地平面的范围都是无限的）。(b) 求任意点 P = (0, y, z) 处的电势和电场。

4.62 导电平面上方的导线沿图 P4.62 所示方向承载电流 I_1 和 I_2。请记住，电流的方向是根据正电荷的移动来定义的，对应于 I_1 和 I_2 的镜像电流的方向是什么？



第5章

静磁场

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Objectives

Upon learning the material presented in this chapter, you should be able to:

1. Calculate the magnetic force on a current-carrying wire placed in a magnetic field and the torque exerted on a current loop.
2. Apply the Biot–Savart law to calculate the magnetic field due to current distribution.
3. Apply Ampère's law to configurations with appropriate symmetry.
4. Explain magnetic hysteresis in ferromagnetic materials.
5. Calculate the inductance of a solenoid, a coaxial transmission line, or other configurations.
6. Relate the magnetic energy stored in a region to the magnetic field distribution in that region.

章节内容

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目标

学习本章介绍的材料后，您应该能够：

1. 计算放置在磁场中的载流导线上的磁力以及施加在电流环上的扭矩。 2. 应用毕奥-萨伐尔定律计算电流分布产生的磁场。 3. 将安培环路定律应用于具有适当对称性的配置。 4. 解释铁磁材料中的磁滞现象。 5. 计算螺线管、同轴传输线或其他配置的电感。 6. 将某个区域中存储的磁能与该区域的磁场分布联系起来。

Overview

This chapter on magnetostatics parallels the preceding one on electrostatics. Stationary charges produce static electric fields, and steady (i.e., non–time-varying) currents produce static magnetic fields. When $\partial/\partial t = 0$, the magnetic fields in a medium with magnetic permeability μ are governed by the second pair of Maxwell's equations (Eqs. (4.3a,b)):

$$\nabla \cdot \mathbf{B} = 0, \tag{5.1a}$$

$$\nabla \times \mathbf{H} = \mathbf{J}, \tag{5.1b}$$

where $\mathbf{J}$ is the current density. The ***magnetic flux density*** $\mathbf{B}$ and the ***magnetic field intensity*** $\mathbf{H}$ are related by

$$\mathbf{B} = \mu \mathbf{H}. \tag{5.2}$$

When examining electric fields in a dielectric medium in Chapter 4, we noted that the relation $\mathbf{D} = \varepsilon \mathbf{E}$ is valid only when the medium is linear and isotropic. These properties, which hold true for most materials, allow us to treat the permittivity ε as a constant, scalar quantity that is independent of both the magnitude and the direction of $\mathbf{E}$. A similar statement applies to the relation given by Eq. (5.2). With the exception of ferromagnetic materials, for which the relationship between $\mathbf{B}$ and $\mathbf{H}$ is nonlinear, most materials are characterized by constant permeabilities.

► Furthermore, $\mu = \mu_0$ for most dielectrics and metals (excluding ferromagnetic materials). ◄

The objectives of this chapter are to develop an understanding of the relationship between steady currents and the magnetic flux $\mathbf{B}$ and field $\mathbf{H}$ due to various types of current distributions and in various types of media and to introduce a number of related quantities, such as the magnetic vector potential $\mathbf{A}$, the magnetic energy density w_m, and the inductance of a conducting structure, L. The parallelism that exists between these magnetostatic quantities and their electrostatic counterparts is elucidated in **Table 5-1**.

5-1 Magnetic Forces and Torques

The electric field $\mathbf{E}$ at a point in space has been defined as the electric force $\mathbf{F}_e$ per unit charge acting on a charged test particle placed at that point. We now define the ***magnetic flux density*** $\mathbf{B}$ at a point in space in terms of the ***magnetic force*** $\mathbf{F}_m$ that acts on a charged test particle moving with velocity $\mathbf{u}$ through that point. Experiments revealed that a particle of charge q moving with velocity $\mathbf{u}$ in a magnetic field experiences a magnetic force $\mathbf{F}_m$ given by

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} \qquad \text{(N)}. \tag{5.3}$$

Accordingly, the strength of $\mathbf{B}$ is measured in newtons/(C·m/s), which is also called the tesla (T). For a positively charged particle, the direction of $\mathbf{F}_m$ is that of the cross product $\mathbf{u} \times \mathbf{B}$, which is perpendicular to the plane containing $\mathbf{u}$ and $\mathbf{B}$ and governed by the right-hand rule. If q is negative, the direction of $\mathbf{F}_m$ is reversed (**Fig. 5-1**). The magnitude of $\mathbf{F}_m$ is given by

$$F_m = quB \sin\theta, \tag{5.4}$$

where θ is the angle between $\mathbf{u}$ and $\mathbf{B}$.

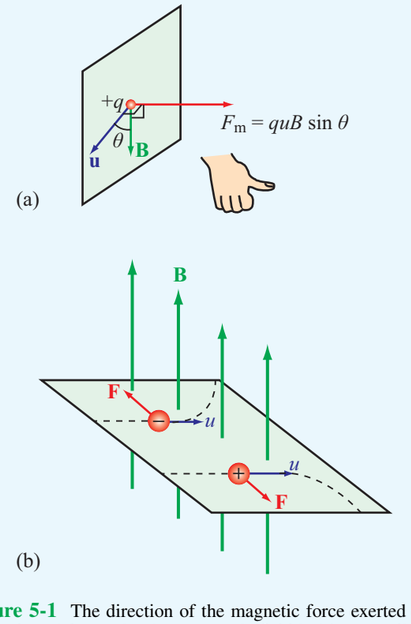


Figure 5-1 The direction of the magnetic force exerted on a charged particle moving in a magnetic field is (a) perpendicular to both $\mathbf{B}$ and $\mathbf{u}$ and (b) depends on the charge polarity (positive or negative).

► We note that F_m is maximum when $\mathbf{u}$ is perpendicular to $\mathbf{B}$ ($\theta = 90°$) and zero when $\mathbf{u}$ is parallel to $\mathbf{B}$ ($\theta = 0$ or 180°). ◄

概述

本章关于静磁学的内容与前一章关于静电学的内容类似。静电荷产生静电场，稳定（即非时变）电流产生静磁场。当 ∂/∂t = 0 时，磁导率为 μ 的介质中的磁场由第二对麦克斯韦方程组控制（方程 (4.3a,b)）：

$$\nabla \cdot \mathbf{B} = 0, \tag{5.1a}$$

$$\nabla \times \mathbf{H} = \mathbf{J}, \tag{5.1b}$$

其中 J 是电流密度。磁通密度 B 和磁场强度 H 的关系为

$$\mathbf{B} = \mu \mathbf{H}. \tag{5.2}$$

在第 4 章研究介电介质中的电场时，我们注意到关系式 D = ε E 仅当介质是线性且各向同性时才有效。这些对于大多数材料都适用的属性使我们能够将介电常数 ε 视为一个恒定的标量，它与 E 的大小和方向无关。类似的陈述适用于方程 1 给出的关系。 （5.2）。除了铁磁材料（B 和 H 之间的关系是非线性的）之外，大多数材料都具有恒定磁导率的特征。

► 此外，μ = μ0 对于大多数电介质和金属（不包括铁磁材料）。 ◄

本章的目标是加深对由于各种类型的电流分布和各种介质中的稳定电流与磁通量 B 和磁场 H 之间的关系的理解，并引入许多相关量，例如磁矢量势 A、磁能密度 wm 和导电结构的电感 L。表 5-1 阐明了这些静磁量与其静电对应量之间存在的并行性。

5-1 磁力和扭矩

空间中某一点的电场 E 被定义为作用于放置在该点的带电测试粒子上的每单位电荷的电力 Fe。现在，我们用作用在带电测试粒子上的磁力 Fm 来定义空间中某一点的磁通密度 B

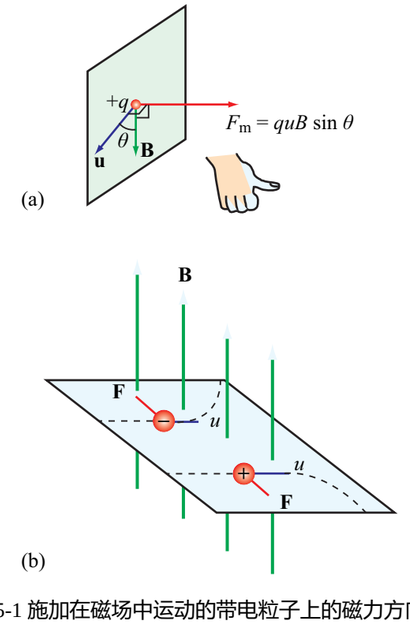


图 5-1 施加在磁场中运动的带电粒子上的磁力方向 (a) 垂直于 B 和 u，(b) 取决于电荷极性（正或负）。

通过该点的速度 u。实验表明，在磁场中以速度 u 运动的电荷 q 粒子会受到由下式给出的磁力 Fm

$$\mathbf{F}_m = q\mathbf{u} \times \mathbf{B} \qquad \text{(N)}. \tag{5.3}$$

因此，B 的强度以牛顿/ (C·m/s) 为单位，也称为特斯拉 (T)。对于带正电的粒子，Fm 的方向是叉积 u × B 的方向，垂直于包含 u 和 B 的平面并受右手定则约束。如果q为负，则Fm的方向相反（图5-1）。 Fm 的大小由下式给出

$$F_m = quB \sin\theta, \tag{5.4}$$

其中 θ 是 u 和 B 之间的角度。

► 我们注意到 Fm当 u 垂直于 B 时最大（θ = 90°）且当 u 平行于 B (θ =0 或 180°). ◄

Table 5-1 Attributes of electrostatics and magnetostatics.

Attribute	Electrostatics	Magnetostatics
Sources	Stationary charges ρ_v	Steady currents $\mathbf{J}$
Fields and Fluxes	$\mathbf{E}$ and $\mathbf{D}$	$\mathbf{H}$ and $\mathbf{B}$
Constitutive parameter(s)	ε and σ	μ
Governing equations		
• Differential form	$\nabla\cdot\mathbf{D}=\rho_v$ $\nabla\times\mathbf{E}=0$	$\nabla\cdot\mathbf{B}=0$ $\nabla\times\mathbf{H}=\mathbf{J}$
• Integral form	$\oint_S\mathbf{D}\cdot d\mathbf{s}=Q$ $\oint_C\mathbf{E}\cdot d\mathbf{l}=0$	$\oint_S\mathbf{B}\cdot d\mathbf{s}=0$ $\oint_C\mathbf{H}\cdot d\mathbf{l}=I$
Potential	Scalar V, with $\mathbf{E}=-\nabla V$	Vector $\mathbf{A}$, with $\mathbf{B}=\nabla\times\mathbf{A}$
Energy density	$w_e=\frac{1}{2}\varepsilon E^2$	$w_m=\frac{1}{2}\mu H^2$
Force on charge q	$\mathbf{F}_e=q\mathbf{E}$	$\mathbf{F}_m=q\mathbf{u}\times\mathbf{B}$
Circuit element(s)	C and R	L

If a charged particle resides in the presence of both an electric field $\mathbf{E}$ and a magnetic field $\mathbf{B}$, then the total ***electromagnetic force*** acting on it is

$$\mathbf{F}=\mathbf{F}_e+\mathbf{F}_m=q\mathbf{E}+q\mathbf{u}\times\mathbf{B}=q(\mathbf{E}+\mathbf{u}\times\mathbf{B}). \tag{5.5}$$

The force expressed by Eq. (5.5) also is known as the ***Lorentz force***. Electric and magnetic forces exhibit a number of important differences:

1. Whereas the electric force is always in the direction of the electric field, the magnetic force is always perpendicular to the magnetic field.
2. Whereas the electric force acts on a charged particle whether or not it is moving, the magnetic force acts on it only when it is in motion.
3. Whereas the electric force expends energy in displacing a charged particle, the magnetic force does no work when a particle is displaced.

This last statement requires further elaboration. Because the magnetic force $\mathbf{F}_m$ is always perpendicular to $\mathbf{u}$, $\mathbf{F}_m\cdot\mathbf{u}=0$. Hence, the work performed when a particle is displaced by a differential distance $d\mathbf{l}=\mathbf{u}\,dt$ is

$$dW=\mathbf{F}_m\cdot d\mathbf{l}=(\mathbf{F}_m\cdot\mathbf{u})\,dt=0. \tag{5.6}$$

► Since no work is done, a magnetic field cannot change the kinetic energy of a charged particle; the magnetic field can change the direction of motion of a charged particle, but not its speed. ◄

Exercise 5-1: An electron moving in the positive x direction perpendicular to a magnetic field is deflected in the negative z direction. What is the direction of the magnetic field?

Answer: Positive y direction. (See EM.)

Exercise 5-2: A proton moving with a speed of 2×10^6 m/s through a magnetic field with magnetic flux density of 2.5 T experiences a magnetic force of magnitude 4×10^{-13} N. What is the angle between the magnetic field and the proton's velocity?

5-1 MAGNETIC FORCES AND TORQUES 229

表 5-1 静电学和静磁学的属性。

Attribute	Electrostatics	Magnetostatics
Sources	Stationary charges ρ_v	Steady currents $\mathbf{J}$
Fields and Fluxes	$\mathbf{E}$ and $\mathbf{D}$	$\mathbf{H}$ and $\mathbf{B}$
Constitutive parameter(s)	ε and σ	μ
Governing equations		
• Differential form	$\nabla\cdot\mathbf{D}=\rho_v$ $\nabla\times\mathbf{E}=0$	$\nabla\cdot\mathbf{B}=0$ $\nabla\times\mathbf{H}=\mathbf{J}$
• Integral form	$\oint_S\mathbf{D}\cdot d\mathbf{s}=Q$ $\oint_C\mathbf{E}\cdot d\mathbf{l}=0$	$\oint_S\mathbf{B}\cdot d\mathbf{s}=0$ $\oint_C\mathbf{H}\cdot d\mathbf{l}=I$
Potential	Scalar V, with $\mathbf{E}=-\nabla V$	Vector $\mathbf{A}$, with $\mathbf{B}=\nabla\times\mathbf{A}$
Energy density	$w_e=\frac{1}{2}\varepsilon E^2$	$w_m=\frac{1}{2}\mu H^2$
Force on charge q	$\mathbf{F}_e=q\mathbf{E}$	$\mathbf{F}_m=q\mathbf{u}\times\mathbf{B}$
Circuit element(s)	C and R	L

如果带电粒子同时存在电场 E 和磁场 B，则作用在其上的总电磁力为

$$\mathbf{F}=\mathbf{F}_e+\mathbf{F}_m=q\mathbf{E}+q\mathbf{u}\times\mathbf{B}=q(\mathbf{E}+\mathbf{u}\times\mathbf{B}). \tag{5.5}$$

力由方程式表示。 (5.5) 也称为洛伦兹力。电力和磁力表现出许多重要的差异：

1. 电力始终沿电场方向，而磁力始终垂直于磁场方向。
2. 无论带电粒子是否运动，电力都会作用在带电粒子上，而磁力仅在带电粒子运动时才作用在带电粒子上。 3. 电力在移动带电粒子时会消耗能量，而磁力在粒子移动时不起作用。

最后的陈述需要进一步阐述。因为磁力 $\mathbf{F}_m$ 始终垂直于 u，所以 $\mathbf{F}_m\cdot\mathbf{u}=0$。

因此，当粒子位移差分距离 $d\mathbf{l}=\mathbf{u}\,dt$ 时所做的功为

$$dW=\mathbf{F}_m\cdot d\mathbf{l}=(\mathbf{F}_m\cdot\mathbf{u})\,dt=0. \tag{5.6}$$

► 由于没有做功，磁场不能改变带电粒子的动能；磁场可以改变带电粒子的运动方向，但不能改变其速度。 ◄

练习 5-1：在垂直于磁场的正 x 方向上移动的电子会在负 z 方向上偏转。磁场的方向是什么？

答案：y 轴正方向。 （参见EM。）

练习 5-2：质子以 2×10^6 m/s 的速度穿过磁通密度为 2.5 T 的磁场时，会受到强度为 4×10^{-13} N 的磁力。磁场与质子速度之间的角度是多少？

Answer: $\theta = 30°$ or $150°$. (See EM.)

Exercise 5-3: A charged particle with velocity $\mathbf{u}$ is moving in a medium with uniform fields $\mathbf{E} = \hat{\mathbf{x}}E$ and $\mathbf{B} = \hat{\mathbf{y}}B$. What should $\mathbf{u}$ be so that the particle experiences no net force?

Answer: $\mathbf{u} = \hat{\mathbf{z}}E/B$. ($\mathbf{u}$ may also have an arbitrary y component u_y.) (See EM.)

5-1.1 Magnetic Force on a Current-Carrying Conductor

A current flowing through a conducting wire consists of charged particles *drifting* through the material of the wire. Consequently, when a current-carrying wire is placed in a magnetic field, it experiences a force equal to the sum of the magnetic forces acting on the charged particles moving within it. Consider, for example, the arrangement shown in **Fig. 5-2** in which a vertical wire oriented along the z direction is placed in a magnetic field $\mathbf{B}$ (produced by a magnet) oriented along the $-\hat{\mathbf{x}}$ direction (into the page). With no current flowing in the wire, $\mathbf{F}_m = 0$ and the wire maintains its vertical orientation (**Fig. 5-2(a)**), but when a current is introduced in the wire, the wire deflects to the left ($-\hat{\mathbf{y}}$ direction) if the current direction is upward ($+\hat{\mathbf{z}}$ direction) and to the right ($+\hat{\mathbf{y}}$ direction) if the current direction is downward ($-\hat{\mathbf{z}}$ direction). The directions of these deflections are in accordance with the cross product given by Eq. (5.3).

To quantify the relationship between $\mathbf{F}_m$ and the current I flowing in a wire, let us consider a small segment of the wire of cross-sectional area A and differential length $d\mathbf{l}$ with the direction of $d\mathbf{l}$ denoting the direction of the current. Without loss of generality, we assume that the charge carriers constituting the current I are exclusively electrons, which is always a valid assumption for a good conductor. If the wire contains a free-electron charge density $\rho_{ve} = -N_e e$, where N_e is the number of moving electrons per unit volume, then the total amount of moving charge contained in an elemental volume of the wire is

$$dQ = \rho_{ve} A\, dl = -N_e e A\, dl, \tag{5.7}$$

and the corresponding magnetic force acting on dQ in the presence of a magnetic field $\mathbf{B}$ is

$$d\mathbf{F}_m = dQ\, \mathbf{u}_e \times \mathbf{B} = -N_e e A\, dl\, \mathbf{u}_e \times \mathbf{B}, \tag{5.8a}$$

where $\mathbf{u}_e$ is the drift velocity of the electrons. Since the direction of a current is defined as the direction of flow of positive charges, the electron drift velocity $\mathbf{u}_e$ is parallel to $d\mathbf{l}$, but opposite in direction. Thus, $dl\, \mathbf{u}_e = -d\mathbf{l}\, u_e$ and Eq. (5.8a) becomes

$$d\mathbf{F}_m = N_e e A u_e\, d\mathbf{l} \times \mathbf{B}. \tag{5.8b}$$

From Eqs. (4.11) and (4.12), the current I flowing through a cross-sectional area A due to electrons with density $\rho_{ve} = -N_e e$ and moving with velocity $-u_e$ is

$$I = \rho_{ve}(-u_e)A = (-N_e e)(-u_e)A = N_e e A u_e.$$

Hence, Eq. (5.8b) may be written in the compact form

$$d\mathbf{F}_m = I\, d\mathbf{l} \times \mathbf{B} \qquad \text{(N)}. \tag{5.9}$$

For a closed circuit of contour C carrying a current I, the total magnetic force is

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \qquad \text{(N)}. \tag{5.10}$$

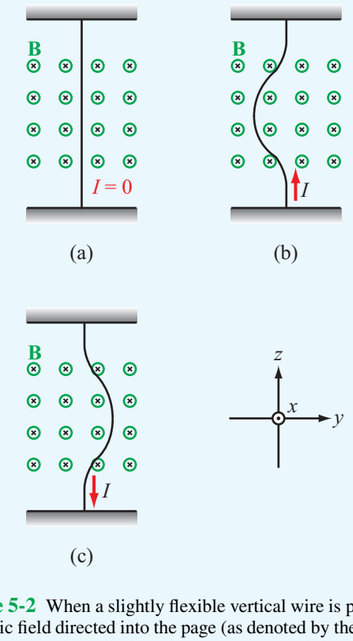


Figure 5-2 When a slightly flexible vertical wire is placed in a magnetic field directed into the page (as denoted by the crosses), it is (a) not deflected when the current through it is zero, (b) deflected to the left when I is upward, and (c) deflected to the right when I is downward.

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答案：θ = 30° 或 150°。 （参见EM。）

练习 5-3：速度为 u 的带电粒子在均匀场 E = x̂E 和 B = ŷB 的介质中运动。你应该怎样才能使粒子不受净力的影响？

答案：u = ẑE/B。 （u 也可以有任意 y 分量 u_y。）（参见 EM。）

5-1.1 载流导体上的磁力

流经导线的电流由漂移穿过导线材料的带电粒子组成。因此，当载流导线放置在磁场中时，它所受到的力等于作用在其中移动的带电粒子上的磁力之和。例如，考虑图 5-2 中所示的布置，其中沿 z 方向定向的垂直导线放置在沿 $-\hat{x}$ 方向（进入页面）定向的磁场 B（由磁铁产生）中。当导线中没有电流流动时，$F_m = 0$ 且导线保持垂直方向（图 5-2（a））， 但当导线中引入电流时，如果电流方向向上（$+\hat{z}$ 方向），则导线向左偏转（$-\hat{y}$ 方向）；如果电流方向向下（$-\hat{z}$ 方向），导线向右偏转（$+\hat{y}$ 方向）。这些偏转的方向与方程式给出的叉积一致。 （5.3）。

为了量化 F_m 与导线中流动的电流 I 之间的关系，让我们考虑一小段横截面积为 A 且长度微分为 dl 的导线，其中 dl 的方向表示电流方向。不失一般性，我们假设构成电流 I 的电荷载流子完全是电子，这对于良导体始终是有效的假设。如果导线包含自由电子电荷密度 $\rho_{ve} = -N_e e$，其中 N_e 是每单位体积移动电子的数量，则导线元素体积中包含的移动电荷总量为

$$dQ = \rho_{ve} A\, dl = -N_e e A\, dl, \tag{5.7}$$

在磁场 B 存在的情况下，作用在 dQ 上的相应磁力为

$$d\mathbf{F}_m = dQ\, \mathbf{u}_e \times \mathbf{B} = -N_e e A\, dl\, \mathbf{u}_e \times \mathbf{B}, \tag{5.8a}$$

其中 u_e 是电子的漂移速度。由于电流方向定义为正电荷流动方向，因此电子漂移速度 u_e 平行于 dl，但方向相反。因此，$dl\, u_e = -dl\, u_e$ 和方程。 (5.8a) 变为

$$d\mathbf{F}_m = N_e e A u_e\, d\mathbf{l} \times \mathbf{B}. \tag{5.8b}$$

从方程式。 (4.11) 和 (4.12) 中，密度为 $\rho_{ve} = -N_e e$ 且以速度 $-u_e$ 运动的电子流过横截面积 A 的电流 I 为

$$I = \rho_{ve}(-u_e)A = (-N_e e)(-u_e)A = N_e e A u_e.$$

因此，等式。 (5.8b) 可以写成紧凑形式

$$d\mathbf{F}_m = I\, d\mathbf{l} \times \mathbf{B} \qquad \text{(N)}. \tag{5.9}$$

对于承载电流 I 的轮廓 C 的闭合电路，总磁力为

$$\mathbf{F}_m = I \oint_C d\mathbf{l} \times \mathbf{B} \qquad \text{(N)}. \tag{5.10}$$

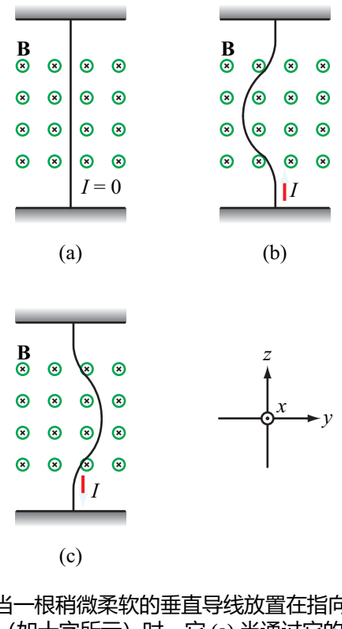


图 5-2 当一根稍微柔软的垂直导线放置在指向页面的磁场中（如十字所示）时，它 (a) 当通过它的电流为零时不偏转，(b) 当 I 向上时向左偏转，以及 (c) 当 I 向下时向右偏转。